LUIS GOMEZ

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ABOUT ME

Mathematics PhD candidate seeking to work with large-scale data, applying mathematical and statistical methods to understand its structure.

EDUCATION

Doctor of Philosophy, Mathematics | University of Iowa, Iowa City, IA | Expected Dec 2028

Master of Science, Mathematics | University of Iowa, Iowa City, IA | Aug 2023 – Dec 2025

Bachelor of Science, Mathematics | University of Arkansas, Fayetteville, AR | May 2020 – May 2023

TECHNICAL SKILLS

Languages: Python, R, Bash, LaTeX

Machine learning & statistics: scikit-learn, Keras, random forests, gradient boosting, regression, ensemble and forecast-combination methods

Data at scale: pandas, NumPy, PySpark, PyArrow / Parquet, distributed computing on HPC clusters

Visualization & tooling: Matplotlib, Seaborn, Git / GitHub, pytest, Jupyter

DATA AND MACHINE LEARNING PROJECTS

- **Options Pricing Engine (Python):** Developed an open-source framework to price European options, featuring five interoperable engines: Black-Scholes with analytical Greeks, a CRR binomial tree, Monte Carlo simulations with antithetic and control variates, a Crank-Nicolson PDE solver, and Heston stochastic volatility models.
- **Equity Premium Forecasting Reproduction (R):** Reproduced the Rapach, Strauss & Zhou (2010) out-of-sample framework - expanding-window OLS across 15 macroeconomic predictors, five combination schemes (mean, median, trimmed mean, DMSPE at two discount factors), evaluated against the historical-average benchmark. Matched the paper's Table 1 for 14/15 predictors; combination forecasts delivered R^2_OS of 3.6%.
- **Probabilistic Recognition of Almost Primes (Python, PySpark):** Built distributed ML pipelines on an HPC cluster to classify integers up to 2^176 in magnitude by prime-factor count over datasets of 2-4M integers. Random forests on modular-residue encodings reached 0.65 AUC vs. 0.50 baseline, with all predictive power coming from divisibility by small primes, indicating the model recovered no structure that cheap arithmetic does not already reveal.

TEACHING EXPERIENCE

Instructor | University of Iowa, Iowa City, IA | Aug 2025 – May 2026

- College Algebra
- Lectured 30+ students on algebraic equations, inequalities, functions, and graphing techniques.

Graduate Teaching Assistant | University of Iowa, Iowa City, IA | Aug 2023 – May 2025

- Quantitative Reasoning for Business and Calculus for the Biological Sciences
- Led discussion sections of 30–160 students, answering questions and clarifying concepts.

RESEARCH EXPERIENCE

University of Arkansas, Fayetteville, AR | Oct 2021 – Dec 2022

- Computed minimal Gröbner bases in Macaulay2 across generated families of ideals, extending published 2×2 determinant methods to the harder 3×3 determinant case.
- Examined the resulting output for a recognizable structural pattern that would characterize the solution set, following the geometric configuration that made the 2×2 case tractable.

Mathematical Sciences Research Institute, Berkeley, CA | Jun 2022 – Jul 2022

- Use SageMath to convert neural codes into their associated monomial ideals and search the generators for algebraic signatures which certify a code admits no convex realization.
- Used the computational evidence to narrow where the equivalence held and where it broke down, guiding the direction for a proof.

Rochester Institute of Technology, Rochester, NY | Jun 2021 – Jul 2021

- Identified recurring structural features that reliably predict a graph invariant, allowing us to completely classify families of graphs by their Zero Forcing number.
- Generalized from computed cases to a closed-form expression for the invariant on circulant graphs, turning an observed pattern across small instances into a formula valid for all n.

PUBLICATIONS

"Probabilistic Recognition of Almost Primes by Machine Learning" with Y. Ye (in preparation).

"Failed Zero Forcing Number of Trees and Circulant Graphs" with K. Rubi, J. Terrazas, D.A. Narayan, R. Flórez, Theory and Applications of Graphs (2025).

"All Graphs with a Failed Zero Forcing Number of Two" with K. Rubi, J. Terrazas, D.A. Narayan, Symmetry, 13, 2221, 2021.

TALKS

Representation Theory Seminar, University of Iowa, Apr 10, 2026: "Market hierarchies as p-adic spaces"

Representation Theory Seminar, University of Iowa, Mar 11, 2026: “Fourier Analysis on Number Fields”

Representation Theory Seminar, University of Iowa, Feb 20 & 27, 2026: "Automorphic representations and L-functions for GL(1) over Q"

Joint Mathematics Meetings, Jan 4, 2023: "Wheel Signatures of Neural Codes"

Joint Mathematics Meetings, Apr 8, 2022: "Failed Zero Forcing Graphs"

AWARDS & CERTIFICATES

Fellowships & Awards: Graduate Assistance in Areas of National Need (GAANN) Fellowship (Fall 2026-Spring 2027); Kleinfeld Endowment Award (Fall 2023).

Certifications: Bloomberg Finance Fundamentals; Center for the Integration of Research, Teaching and Learning (CIRTL) Practitioner level, UIOWA Bridge program.

DataCamp: Supervised Learning with scikit-learn, Unsupervised Learning in Python, Quantitative Risk Management in Python, GARCH Models in Python.